

Soumyajoy Kundu

Curriculum Vitae

Sedin Technologies
Chennai, India
☎ +91-7980585466
✉ sk.cmids25@gmail.com
🔗 soumyajoykundu
🌐 Soumyajoy Kundu

Education

- Aug 2023 – **Masters in Data Science**, *Chennai Mathematical Institute (CMI)*.
Jul 2025
 - GPA: 9.38 / 10.00
 - Ranked 5th in Department
- Oct 2020 – **Bachelors (Honours) in Statistics**, *Maulana Azad College, University of Calcutta*.
Aug 2023
 - GPA: 9.163 / 10.000
 - Ranked 1st in Department and University

Work Experience

- Jun 2025 – **Junior Data Scientist**, *Dataculture*, Sedin Technologies, Chennai.
Present
 - Developing multimodal AI systems integrating vision, language and audio models for contextual reasoning and enterprise decision support.
 - Designed and deployed GPT-4o powered conversational and document understanding systems integrated with Salesforce and Microsoft Teams.
- Nov 2025 – **Research Assistant**, *ForeML Lab*, Sorbonne University and Sorbonne Center for AI.
Present
 - Working on robust demand forecasting in diffusion-driven environments, extending classical Bass-type models to incorporate stochastic perturbations and structural uncertainty.
 - Investigating statistical and machine learning approaches for parameter stability and long-horizon forecasting under the supervision of *Dr. Tanujit Chakraborty*.
- Jan 2025 – **Research Intern**, *Theia Technologies*, Bangalore.
Apr 2025
 - Built real-time object tracking systems using SAM2 and SAMURAI architectures under dynamic camera motion.
 - Deployed PTZ-controlled tracking pipelines on RTSP streams, improving robustness under jitter and rapid object displacement.
- Aug 2024 – **Teaching Assistant**, *Chennai Mathematical Institute*, Chennai.
Apr 2025
 - Assisted in courses on Probability and Statistics with R, Visualization and Linear Algebra and its Applications.
 - Conducted tutorial sessions and supported coursework involving statistical modeling, matrix methods and data analysis.
- May 2024 – **Research Associate**, *Reserve Bank of India*, Chennai.
Jul 2024
 - Studied the impact of RBI liquidity management operations on USD/INR exchange rate dynamics using macro-financial data.
 - Implemented econometric models including VAR, ARDL, ECM and GARCH, demonstrating statistical efficiency over RNN/LSTM models in small-sample settings.

Key Projects

- 2024 **Optimizing Acrobot Dynamics via Deep Reinforcement Learning,** 
Supervisor: *Pranabendu Misra*, Assistant Professor, Chennai Mathematical Institute
- Implemented a Deep Q-Network (DQN) to learn optimal control policies in the Acrobot environment, modeling the problem as a Markov Decision Process.
 - Performed systematic hyperparameter tuning (learning rate, discount factor, epsilon decay) to stabilize training and achieve consistent reward convergence.
- 2025 **Multi-View 3D Reconstruction via Bundle Adjustment,** 
Supervisor: *Kavita Sutar Deshpande*, Visiting Professor, Chennai Mathematical Institute
- Developed a complete Structure-from-Motion pipeline including feature extraction, pose estimation, triangulation, and non-linear optimization.
 - Implemented Levenberg–Marquardt based bundle adjustment to jointly refine camera parameters and 3D structure by minimizing global reprojection error.
- 2026 **KovaiDelivery: Multi-Agent Autonomous Drone Logistics,** 
 - Formulated large-scale drone fleet coordination (100+ agents) as a constrained optimization problem under dynamic weather and battery constraints, designing safety-aware allocation policies.
 - Implemented typed inter-agent communication and stateful decision pipelines, ensuring robust and failure-resistant multi-agent coordination.
- 2024 **Data-driven Graph Construction: Unlocking Insights through Clustering and Spectral Partitioning,** 
Supervisor: *Priyavrat Deshpande*, Associate Professor, Chennai Mathematical Institute
- Constructed similarity graphs from structured datasets and analyzed graph Laplacian properties to study clustering behavior under non-convex geometries.
 - Implemented spectral partitioning techniques using Fiedler vectors and k-means, demonstrating improved partition stability compared to classical distance-based clustering methods.
- 2025 **GeomVLM: Lightweight Vision–Language Model,** 
 - Designed a compact multimodal architecture combining CNN-based visual encoders with Transformer text encoders and trained using CLIP-style contrastive objectives.
 - Constructed a shared embedding space for structured image–text alignment and evaluated cross-modal retrieval performance using synthetic geometric datasets.

Publications

- 2025 **Ozempic, Mounjaro, and other GLP-1 Drugs: How Effective Are They for Weight Loss?**, *Datakulture*, Sedin Technologies, 
- Co-authored a data-driven analysis of GLP-1 receptor agonists (Semaglutide, Tirzepatide), synthesizing findings from clinical trials and real-world studies to evaluate weight-loss efficacy across patient cohorts and dosage variations.
 - Examined benefit-risk trade-offs including metabolic improvements, gastrointestinal side effects, long-term sustainability, and cohort-specific response patterns based on published medical evidence.

Paper and Poster Presentations

- 2024 **4th International Conference on Mathematical, Engineering and Management Sciences** organised by NIT Meghalaya & NIT Patna.
- Presented a *paper* on "A Statistical Analysis of the Paradigm Shift in Consumer Behavior in the Digital Era" demonstrating methodology and key findings of the research which was later selected for journal proceedings.

- 2024 **Conference for Statistics and Programming in Clinical Research** organised by Indian Association for Statistics in Clinical Trials (**IASCT**).
- Presented a *poster* on "*Data-driven Graph Construction: Unlocking Insights through Clustering and Spectral Partitioning*" demonstrating methodology and key findings of the research.

Conferences, Workshops and Certifications

- 2025 **International Conference on Applied AI and Scientific Machine Learning (**CASML**)**, Indian Institute of Science, Bangalore
- 2025 **Winter School on Deep Learning: Reinforcing Science with Foundation Models (**WSDL**)**, Electronics and Communication Sciences Unit, Indian Statistical Institute

Invited Talks and Seminars

- 2025 **Seminar on National Mathematics Day: “*Probabilistic Modeling*”**, Department of Statistics, Maulana Azad College
- 2024 **Seminar: “*Data-driven Graph Construction*”**, Data Science Seminar, CMI
- 2024 **Invited Talk: “*Conformal Prediction*”**, Top 6 Speakers across India, The AI Hour, Vizura AI Labs, Vizura Technologies Pvt. Ltd. 
- 2024 **Invited Talk: “*AI for Everyone*”**, Department of Statistics, Maulana Azad College

Achievements

- 2026 **Qualified Council of Scientific and Industrial Research National Eligibility Test**
- All India Rank 18 (CSIR-NET JRF)
 - Percentile : 99.93
- 2025 **Rising Star of the Quarter (Q3 2025)**, Datakulture, Sedin Technologies
- 2025 **Qualified Graduate Aptitude Test in Engineering (GATE)**
- **Statistics** : All India Rank 03
 - **Data Science and Artificial Intelligence** : All India Rank 129
- 2023 & 2024 **Qualified Joint Admission Test for Masters (IIT JAM)**, All India Rank 100 & 11
- 2023 & 2024 **Qualified Masters in Statistics entrance exam by Indian Statistical Institute**, All India Rank 33 & 27
- 2023 **Qualified Masters in Mathematics entrance exam by Indian Institutes of Science Education and Research, Kolkata**, Top 12 across India
- 2023 **Qualified Common University Entrance Test - Postgraduate in Statistics**, All India Rank 01
- 2023 **Secured 1st Rank in the University for overall academic performance in Undergraduate Degree**, University of Calcutta

Relevant Coursework

- Mathematics* MATHEMATICAL ANALYSIS AND METHODS, LINEAR ALGEBRA AND ITS APPLICATIONS, OPERATIONS RESEARCH, MATHEMATICAL FINANCE
- Probability* PROBABILITY THEORY, STOCHASTIC PROCESSES

Statistics STATISTICAL INFERENCE, LINEAR MODELS, NON-PARAMETRIC STATISTICS, LARGE SAMPLE THEORY, SAMPLE SURVEY, REGRESSION TECHNIQUES, SURVIVAL ANALYSIS, TIME SERIES, MULTIVARIATE ANALYSIS

AI DATA MINING AND MACHINE LEARNING, DEEP LEARNING, REINFORCEMENT LEARNING, NATURAL LANGUAGE PROCESSING, COMPUTER VISION

Computing PROGRAMMING AND DATA STRUCTURES USING PYTHON, NUMERICAL ANALYSIS, DESIGN AND ANALYSIS OF ALGORITHMS, MONTE CARLO METHODS, DISTRIBUTED COMPUTING AND BIG DATA

Technical Skills

Programming Python, R, SQL, Java

ML/DL PyTorch, TensorFlow, Scikit-learn, OpenCV, Nltk, spaCy

Data Pandas, NumPy, ggplot2, PowerBI, E-Views

Other L^AT_EX, Git, Docker, MongoDB

References

- 1 *Dr. Sourish Das*, Associate Professor, Chennai Mathematical Institute, sourish@cmi.ac.in
- 2 *Dr. Sayantan Banerjee*, Associate Professor, Indian Institute of Management Indore, sayantanb@iimdr.ac.in
- 3 *Dr. Kavita Sutar-Deshpande*, Adjunct Professor, Chennai Mathematical Institute, ksutar@cmi.ac.in

Required Information

- 1 **Date of Birth** : 07 January, 2003
- 2 **Country of Passport** : India
- 3 **Residency** : 2E Rajanigandha Apartment beside Mouchak Club, Raja Ram Chand Ghat Road, Panihati, Kolkata - 700114, West Bengal, India.